

Name:	
ID:	
Date:	

Bachelor of Science in Computer Engineering

Worksheet

Computer Engineering Requirements (at least 50 credits)

Computer Engineering Core (32 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Reg, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CE 201	Linear Circuit Analysis I	3	Pre: ENGR 131, MA 166 (min grade of C-), PHYS 172 ConP: MA 261	Lect.: 3h			
☐	CE 207	Electronic Measurement Techniques Lab	1	ConP: CE 201	Lab.: 3h			
☐	CE 202	Linear Circuit Analysis II	3	Pre: CE 201 (min grade of C) ConP: MA 266	Lect.: 3h			
☐	CE 270	Introduction to Digital System Design	4	ConP: CE 201	Lect.: 3h; Lab.: 3h			
☐	CE 255	Introduction to Electronic Analysis and Design	3	Pre: CE 201 (min grade of C)	Lect.: 3h			
☐	CE 208	Electronic Devices and Design Laboratory	1	Pre: CE 207 ConP: CE 255	Lab.: 3h			
☐	CE 301	Signals and Systems	3	Pre: CE 202 (min grade of C), MA 266	Lect.: 3h			
☐	CE 264	Advanced C Programming	3	Pre: CS 159 (min grade of C-)	Lect.: 2h; Lab.: 2h			
☐	CE 362	Microprocessor Systems and Interfacing	4	Pre: EE 270 (min grade of C) or CE 270 (C), CS 159	Lect.: 3h; Lab.: 3h			
☐	CE 368	Data Structures	3	Pre: CE 264 (min grade of C-)	Lect.: 3h			
☐	CE 364	Software Engineering Tools Laboratory	1	Pre: CE 264	Lab.: 3h			
☐	CE 302	Probabilistic Methods in Electrical and Computer Engineering	3	Pre: MA 266 ConP: CE 301	Lect.: 3h			

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Computer Engineering Seminars (1 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CE 200	Electrical and Computer Engineering Sophomore Seminar	0	Sophomore	Lect.: 1h			
☐	CE 400	Professional Development and Career Guidance - Graduation Project I	1	Pre: CE 200, Senior	Lect.: 1h; Lab.: 1h			

Advanced CE Selectives (8 - 10 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CE 337	ASIC Design Laboratory	2	Pre: EE 270 (min grade of C) or CE 270 (min grade of C)	Lect.: 1h; Lab.: 3h			
☐	CE 437	Computer Design and Prototyping	4	Pre: CE 337, CE 362	Lect.: 3h; Lab.: 3h			
☐	CE 468	Introduction to Compilers and Translation Engineering	4	Pre: CE 362, CE 368	Lect.: 3h; Lab.: 2h			
☐	CE 469	Operating System Engineering	4	Pre: CE 368, CE 437	Lect.: 3h; Lab.: 3h			

Senior Design Requirement (4 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CE 477	Digital Systems Senior Project	4	Pre: CE 400 or EE 400, EE Core Curriculum or CE Core Curriculum, Senior	Lect.: 1h; Lab.: 1h			

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Computer Engineering Electives (at least 3 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
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Computer Engineering Credits Planned: _____

Computer Engineering Credits Completed: _____

Computer Engineering Credits Remaining: _____

General Engineering (10 credits)

Introduction to Engineering (7 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CS 159	Programming Applications for Engineers	3	Pre: ENGR 131	Lect.: 3h; Lab.: 2h			

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	ENGR 131	Transforming Ideas to Innovation I	2	-	Lect.: 2h; Lab.: 2h			
☐	ENGR 132	Transforming Ideas to Innovation II	2	Pre: ENGR 131	Lect.: 2h; Lab.: 2h			
OR								
☐	ENGR 100	First-Year Engineering Lectures	1	-	Lect.: 1h			
☐	ENGR 126	Engineering Problem Solving and Computer Tools	3	-	Lect.: 3h			

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Engineering Breadth Requirement (3 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	ENGR 297	Basic Mechanics I (Statics)	3	Pre: PHYS 172 ConP: MA 261	Lect.: 3h			
☐	IE 335	Operation Research – Optimization	3	Pre: MA 265 ConP: EE 302 or CE 302 or IE 332	Lect.: 2h; Lab.: 2h			
☐	IE 336	Operation Research – Stochastic Models	3	Pre: MA 265, IE 230 ConP: EE 302 or CE 302 or IE 332, MA 266	Lect.: 3h			
☐	ME 200	Thermodynamics I	3	Pre: CHM 115 ConP: MA 261, ENGR 132	Lect.: 3h			
☐	MSE 230	Structure and Properties of Materials	3	Pre: CHM 115 (min. grade C-), MA 165 (min grade C-)	Lect.: 3h			
☐	NUCL 200	Introduction to Nuclear Engineering	3	Pre: MA 166 or equivalent, PHYS 172 or equivalent	Lect.: 3h			
☐	NANO 101	Introduction to Nanotechnology	3	Pre: MA 261; MA 266 or MA 262; PHYS 272 or PHYS 241; CHM 115	Lect.: 3h			
☐	CVL 355	Engineering Environmental Sustainability I	3	Pre: Sophomore	Lect.: 3h			

General Engineering Credits Planned: _____

General Engineering Credits Completed: _____

General Engineering Credits Remaining: _____

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Mathematics Requirement (21 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	MA 165	Analytic Geometry and Calculus I	4	Passing Math Placement Test or MAT 110 or MA 158	Lect.: 3h; Lab.: 2h			
☐	MA 166	Analytic Geometry and Calculus II	4	Pre: MA 165	Lect.: 4h			
☐	MA 261	Multivariate Calculus	4	Pre: MA 166	Lect.: 4h			
☐	MA 265	Linear Algebra	3	Pre: MA 166	Lect.: 3h			
☐	MA 266	Ordinary Differential Equations	3	Pre: MA 261	Lect.: 3h			
☐	MA 369	Discrete Mathematics for Computer Engineering	3	Pre: CE 270	Lect.: 3h			

Mathematics Credits Planned: _____
 Mathematics Credits Completed: _____
 Mathematics Credits Remaining: _____

Science Requirement (15-16 credits)

Core (12 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	CHM 115	General Chemistry I	4	ConP: MA 165	Lect.: 3h; Lab.: 2h			
☐	PHYS 172	Modern Mechanics	4	ConP: MA 165	Lect.: 3h; Lab.: 2h			
☐	PHYS 272	Electric and Magnetic Interactions	4	Pre: PHYS 172, ConP: MA 166	Lect.: 3h; Lab.: 2h			

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Science Electives (3-4 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	BIOL 110	Fundamentals of Biology I	4	-	Lect.: 3h; Lab.: 2h			
☐	CHM 116	General Chemistry II	4	Pre: CHM 115	Lect.: 3h; Lab.: 2h			
☐	PHYS 322	Intermediate Optics	3	Pre: PHYS 272 or PHYS 241	Lect.: 3h			
☐	PHYS 342	Modern Physics	3	Pre: PHYS 272 or PHYS 241	Lect.: 3h			
☐	(BIOL 121	Biology I: Diversity, Ecology, and Behavior	2	-	Lect.: 2			
☐	and BIOL 135)	First year Biology Laboratory	2	ConP: BIOL 121, CHM 115	Lab.: 3			
☐	CHM 112	General Chemistry	3	Pre: CHM 115	Lect.: 3h			
☐	EAPS 125	Environmental Science and Conservation	3	-	Lect.: 3h			

Science Credits Planned: _____

Science Credits Completed: _____

Science Credits Remaining: _____

Liberal Arts Requirements (28 credits)

English Language and Communication Skills (10 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
☐	ENGL 100	English for Academic Studies	3	-	Lect.: 3h			
☐	ENGL 106	First-Year Composition	4	Pre: ENGL 100	Lect.: 4h			
☐	COM 114	Fundamentals of Speech Communication	3	Pre: ENGL 100	Lect.: 3h			

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General Education Electives (18 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
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Liberal Arts Credits Planned: _____

Liberal Arts Credits Completed: _____

Liberal Arts Credits Remaining: _____

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Complementary Electives (3-5 credits)

	<u>Course Code</u>	<u>Course Title</u>	<u>CR</u>	<u>Pre-Req, ConP</u>	<u>Contact Hours</u>	<u>Other Information</u>	<u>Semester</u>	<u>Comments</u>
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Complementary Electives Credits Planned: _____

Complementary Electives Credits Complete: _____

Complementary Electives Credits Remaining: _____

Minimum Total Credits Required for Degree: 128

Total Credits Planned: _____

Total Credits Completed: _____

Total Credits Remaining: _____